Project

Deadline: **29 November 2026** Weight: **25%** of the course grade

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General instructions. Work in groups of three to five students. Your submission must contain:

(a) A single **PDF report**, **at most 15 pages** including all figures, and tables. The references and backtest tables (see below) may be placed in an appendix that does not count towards the limit.

(b) The corresponding **code files** in Python (`.py` or `.ipynb`), with a short note on where in the code each table and figure of the report is produced.

The PDF must be fully self-contained: every result, plot and table must be visible in the report without executing the code. A single command or notebook run must regenerate every number and figure from the provided data file alone.

List all group members on the title page.

Project description. This project covers market risk measurement for three individual stocks: VaR and ES estimation by several methods, backtesting those methods, and a short extreme-value analysis of the tail. It uses material from Parts I–VIII.

Data. Use AAPL, META and JPM daily closing prices (adjusted for splits and dividends), 2 January 2013 through 30 June 2026, provided in the file `QRM_2026_Project_prices.csv` (downloaded with `yfinance`; do not download them again, since Yahoo revises adjusted prices back in time). The provided template notebook, `Project_template.ipynb`, reads this file. Define log-returns $R_t = \log(P_t/P_{t-1})$ and the log-loss $L_t = -R_t$ throughout. A forecast for day t uses data up to day $t-1$ only. **Every question is carried out separately for each of the three stocks**. To stay within the page limit, report the three stocks side by side in shared tables, and put per-stock plots in one multi-panel figure. You may use any Python package you find useful.

1. **Stylized facts. [10 points]**

   a) Construct the log-returns of each stock and plot the three series. Comment on where volatility visibly rises and falls together, and where one stock moves on its own.

   b) For each stock, plot the autocorrelation function (ACF) of returns and of *squared* returns for lags $h = 1, \dots, 30$, and apply the Ljung–Box test (Appendix A.4) with $H = 10$ to both series. Explain, using these results, why an i.i.d. model for R_t is a bad description of this data even though the ACF of R_t itself may look close to zero.

   c) QQ-plot each stock's returns against the normal distribution and apply the Jarque and Bera (1980) test (Appendix A.4). State the null hypothesis precisely and interpret the result together with the QQ-plot.

2. **Single-window VaR and ES estimates. [10 points]** For each stock, fix the loss series on its *first* $W = 504$ observations as one estimation window (this is also the window of the first forecast in Question 3). Using the log-loss $L_t = -R_t$, fit and compare, at $\alpha \in \{0.975, 0.99\}$:

a) *Historical simulation*: with $L_{1,W} \geq \cdots \geq L_{W,W}$ the ordered losses and $k = \lfloor W(1-\alpha) \rfloor$,

$$\widehat{\mathrm{VaR}}_\alpha = L_{k+1,W}, \qquad \widehat{\mathrm{ES}}_\alpha = \frac{1}{W(1-\alpha)} \Big(\sum_{i=1}^{k} L_{i,W} + \big(W(1-\alpha) - k\big) L_{k+1,W} \Big).$$

b) *Gaussian*: fit (μ, σ) by sample moments; closed-form VaR/ES.

c) *Student-t*: fit (ν, μ, σ) by maximum likelihood; closed-form VaR/ES. Report $\hat{\nu}$ and discuss what it implies about the tail. Note that σ is a scale parameter, not the standard deviation, and that `scipy.stats.t.fit` can stop at a local optimum.

d) For each stock, overlay the two fitted densities (plus the empirical histogram) in one panel. Where do the three VaR estimates sit relative to each other at each level, and does the ordering match what you would predict from the pdfs in the plot?

The rolling strategy in Questions 3–5. For each stock, produce one-day-ahead forecasts of VaR and ES (VaR only in Question 5) on *every* trading day of the evaluation period, at $\alpha \in \{0.975, 0.99\}$. The evaluation period is the same for every model: all trading days after the first 504 (two years) of the sample, i.e. 2,888 days from 5 January 2015. Each model is estimated on a rolling window of the $W = 504$ days before the forecast date. Historical simulation, Gaussian and Student-t are recomputed every day. The models of Questions 4 and 5 are only re-estimated every 21 trading days (approximately monthly), counting from the first forecast day: on the days between two refit dates, keep the parameters from the most recent refit, but update the forecast with each day's data (for AR-GARCH, advance $\hat{\mu}_t$ and $\hat{\sigma}_t$ through the recursions with the realized returns; for quantile regression, update the features). So you still get, and still backtest, one forecast and one exception indicator per trading day. If a refit fails to converge or ends at the boundary of the parameter space, say what you did and how often.

Combined backtest tables. Collect the VaR backtests of Questions 3–5 in one table per level α, with one row per stock and method (the three methods of Question 3, the two of Question 4 and the quantile regression of Question 5) and the columns: number of exceptions, expected number, hit rate, Kupiec p-value, Christoffersen independence p-value, conditional-coverage p-value, and average pinball loss (Appendix A.1). Collect the ES backtests at $\alpha = 0.975$ in a separate table, with one row per stock and method (the five methods that forecast ES) and the columns: Acerbi–Székely Z, its p-value, and average Fissler–Ziegel score. Fill them in as you go; the comparisons in Questions 3–5 and Question 7 refer to them.

3. **Backtesting. [20 points]** Apply the rolling backtest above to the three methods of Question 2, for each stock.

a) Fill in the rows of these three methods in the combined VaR tables, using the Kupiec (1995) unconditional-coverage test, the Christoffersen (1998) independence and conditional-coverage tests (as in class) and the pinball loss (Appendix A.1).

b) Which methods fail coverage, which fail independence, and at which level? For any case where the Kupiec and conditional-coverage verdicts disagree, say which failure mode is responsible.

c) At $\alpha = 0.975$, apply the Acerbi and Székely (2014) Z test (Appendix A.2) to each method's ES forecasts, with $M = 10000$ simulated samples. Report Z and its simulated p-value in the ES table. Z has no closed-form null distribution: why does its simulated distribution depend on the method being tested?

d) Using the Fissler–Ziegel joint score (Appendix A.3) at $\alpha = 0.975$, rank the three methods by average score for each stock, and run a Diebold and Mariano (1995) test (Appendix A.3) between the best and second-best. State whether the difference is significant, and say precisely what that test does and does not tell you about the two models.

4. **AR-GARCH conditional VaR and ES. [25 points]** For each stock, build the conditional model in two stages, first the mean and then the variance, and then use it inside the rolling scheme. The model is AR(p)–GARCH(1, 1) (Bollerslev, 1986):

$$R_t = \mu_t + \varepsilon_t, \qquad \mu_t = \mu + \sum_{i=1}^{p} \phi_i \, (R_{t-i} - \mu), \qquad \varepsilon_t = \sigma_t Z_t, \qquad \sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2,$$

with Z_t i.i.d. with mean 0 and variance 1 (standard normal or standardized Student-t), $\alpha_0 > 0$, $\alpha_1, \beta_1 \geq 0$ and $\alpha_1 + \beta_1 < 1$. The variance recursion is driven by ε_t, not by R_t. The `arch` package supports autoregressive but not moving-average terms in the mean, so the mean model is restricted to AR(p). State how the variance recursion is initialized.

a) *Mean model.* On the full sample of each stock's returns:

(i) *Choose p.* Fit AR(p) models with constant variance for $p \in \{0, 1, 2, 3\}$ by Gaussian quasi-maximum likelihood, all on the same observations (condition on the first three, e.g. `hold_back=3` in `arch`). Report AIC and BIC for each p and choose p by BIC (Appendix A.4). Do the two criteria agree?

(ii) *Fit.* Report the estimated $\hat{\mu}, \hat{\phi}_1, \ldots, \hat{\phi}_p$ of the chosen model with standard errors. Note that `arch` reports the intercept $c = \mu(1 - \phi_1 - \cdots - \phi_p)$ ("`Const`"), not μ.

(iii) *Check.* Apply the Ljung–Box test with $H = 10$ to the residuals $\hat{\varepsilon}_t$ and to the squared residuals $\hat{\varepsilon}_t^2$ (Appendix A.4 gives the reference distributions). The test on $\hat{\varepsilon}_t$ assumes i.i.d. residuals under the null. Does the result for $\hat{\varepsilon}_t^2$ support that assumption? If not, explain why a rejection on $\hat{\varepsilon}_t$ cannot be taken as evidence that the AR order is wrong.

b) *Variance model.* Keeping the p of (a), fit AR(p)–GARCH(1, 1) on the full sample, once with Gaussian and once with standardized Student-t innovations (`dist="t"` in `arch`). This is a diagnostic fit, used to check the model and compare it with (a), not to forecast. Report the parameter estimates (including $\hat{\nu}$) and compare the two fits by AIC and BIC. For the Gaussian fit, repeat the two Ljung–Box tests of (a)(iii) on the standardized residuals $\hat{z}_t = (R_t - \hat{\mu}_t)/\hat{\sigma}_t$ and on $\hat{z}_t^2$. Did the GARCH step remove the dependence in $\hat{z}_t^2$? If so, is the test on $\hat{z}_t$ now valid, and does it reject? What does this say about your answer to (a)(iii)?

c) *Rolling forecasts and backtest.* With the same p, re-fit both models inside the rolling scheme. With $\hat{\mu}_t, \hat{\sigma}_t$ the conditional mean and volatility of R_t given the data up to $t - 1$, the Gaussian model gives

$$\widehat{\mathrm{VaR}}_{t,\alpha} = -\hat{\mu}_t + \hat{\sigma}_t \Phi^{-1}(\alpha), \qquad \widehat{\mathrm{ES}}_{t,\alpha} = -\hat{\mu}_t + \hat{\sigma}_t \frac{\varphi(\Phi^{-1}(\alpha))}{1 - \alpha},$$

and the Student-t model replaces the standard normal by the unit-variance t_ν distribution: with $q = t_\nu^{-1}(\alpha)$ and $c_\nu = \sqrt{(\nu - 2)/\nu}$,

$$\widehat{\mathrm{VaR}}_{t,\alpha} = -\hat{\mu}_t + \hat{\sigma}_t \, c_\nu \, q, \qquad \widehat{\mathrm{ES}}_{t,\alpha} = -\hat{\mu}_t + \hat{\sigma}_t \, c_\nu \, \frac{g_\nu(q)}{1 - \alpha} \, \frac{\nu + q^2}{\nu - 1},$$

where g_ν is the t_ν density. Backtest both and add their rows to the two tables.

d) Comment on the results. Compared with the unconditional methods of Question 3, does the GARCH filter change the independence p-values? Compared with the Gaussian GARCH model, do Student-t innovations change the Kupiec p-values and the pinball loss? At $\alpha = 0.975$, rank all five methods by Fissler–Ziegel score and run a Diebold–Mariano test between the best unconditional and the best GARCH model. Does the best-ranked model also pass the Z test? Is the answer the same for all three stocks?

5. **Quantile regression. [15 points]** Forecast each stock's VaR directly with a model of the conditional quantile of the loss. The pinball loss (Appendix A.1) is a strictly consistent scoring function for the quantile: in expectation it is minimized by the true VaR, so the lower the average pinball loss, the better the VaR forecast, which is why it appears in the backtest table. If this is the criterion by which VaR forecasts are judged, why not choose the forecast by minimizing it directly? This is what quantile regression does, within a parametric family of forecasts. For each $\alpha \in \{0.975, 0.99\}$, model the conditional α-quantile of L_s given the information up to $s-1$ as a linear function $q_\alpha(x_{s-1}) = \beta_0 + \beta^\top x_{s-1}$ of two features available on day $s-1$: the lagged absolute return $|R_{s-1}|$ and the 20-day trailing sample standard deviation $\hat{\sigma}_{20,s-1}$ of $R_{s-20}, \ldots, R_{s-1}$. At each refit date t, estimate (β_0, β) by minimizing the summed pinball loss over the pairs (x_{s-1}, L_s) of the current window ($W = 504$),

$$(\hat{\beta}_0, \hat{\beta}) = \arg\min_{(\beta_0,\beta)} \sum_{s=t-504}^{t-1} S^{\text{pin}}\big(\beta_0 + \beta^\top x_{s-1},\, L_s\big),$$

dropping the pairs whose features are not yet defined (the first window has 484). Keep $(\hat{\beta}_0, \hat{\beta})$ until the next refit and forecast $\widehat{\text{VaR}}_{t,\alpha} = \hat{\beta}_0 + \hat{\beta}^\top x_{t-1}$. This is the linear quantile regression of Koenker and Bassett (1978) (e.g. `QuantReg` in `statsmodels`, with a constant column). Only VaR is forecast in this question, not ES. Fit one model per stock and level, on that stock's data only.

a) Fit the model inside the rolling scheme and add its rows to the combined table. On how many days is the 99% forecast below the 97.5% one (quantile crossing)? How could you fix it?

b) Using the combined table, does the quantile regression improve on the unconditional methods of Question 3, and on the GARCH models of Question 4? It minimizes the pinball loss in each training window: does it also have the lowest pinball loss out of sample? Compare it with the GARCH model of lower pinball loss by a Diebold–Mariano test on the pinball-loss differences. Is the answer the same for all three stocks?

c) *Model selection and validation* (discussion; no code required). Every choice in this question was fixed for you: the features and the window length. In practice each would have to be chosen from data, e.g. whether to add the lagged returns $R_{t-1}, R_{t-2}, R_{t-3}$. Describe how you would do it without touching the evaluation period: which data you would validate on, how you would split it in time (and why not at random), which loss you would compare, and roughly how many losses beyond the 99% quantile your validation sample, and a 504-day training window, would contain. Finally, audit the other methods in the same way: list the choices behind each (window length, refit frequency, AR order, innovation distribution) and say whether any of them used data from the evaluation period.

6. **Extreme value theory. [10 points]** For each stock, work on the full-sample log-loss series.

a) Question 1 shows that these losses are not i.i.d. In one sentence, say what a GPD fitted to them estimates if the loss series is stationary.

b) Plot the sample mean excess function. Take as threshold u the empirical 95% quantile of the losses (the rule of Part VI), mark it on the plot, and say whether the plot is approximately linear above u. Report the number of exceedances N_u.

c) Fit a GPD $(\hat{\xi}, \hat{\beta})$ to the excesses over u by maximum likelihood (Pickands, 1975; Balkema and de Haan, 1974), and check the fit by comparing the fitted and empirical tail of the excesses. Report $\hat{\xi}$ and say what it implies about the existence of higher moments of the loss distribution. In particular, is $\mathbb{E}[L_t^4]$ finite, as the Ljung–Box test on R_t^2 in Question 1(b) requires?

d) For $x > u$, the tail splits exactly as $\bar{F}(x) = \bar{F}(u)\,\bar{F}_u(x-u)$, with $\bar{F} = 1 - F$ and F_u the excess distribution over u. Estimate $\bar{F}(u)$ by N_u/n and $\bar{F}_u$ by the fitted GPD, and show that this gives the estimators of Appendix A.5 (invert the tail estimate for VaR; use the mean excess function of the GPD for ES). Report $\widehat{\mathrm{VaR}}_{0.999}$ and $\widehat{\mathrm{ES}}_{0.999}$, a level deep enough that historical simulation on this sample has at most three or four exceedances to work with.

e) Extrapolate the Gaussian and Student-t fits of Question 2 (re-fit on the full sample) to the same $\alpha = 0.999$ and compare all three estimates in one table. Which model gives the lowest 99.9% VaR and ES, and by how much, in percent, do the Gaussian and Student-t estimates differ from the GPD estimates?

7. **Recommendation. [10 points]** At most one page. Of the methods backtested in Questions 3–5, which would you put into production for each stock and each level α, and why? Is the answer the same for all three stocks?

References

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A Reference formulas

A.1 Pinball loss

For a VaR forecast r at level α and realized loss y, the pinball loss is

$$S^{\text{pin}}(r, y) = \big(\mathbf{1}\{y \le r\} - \alpha\big)(r - y).$$

If $\mathbb{E}\,|Y| < \infty$, $r \mapsto \mathbb{E}[S^{\text{pin}}(r, Y)]$ is minimized exactly at the α-quantiles of Y (uniquely at $\text{VaR}_\alpha(Y)$ if the quantile is unique): the pinball loss is a strictly consistent scoring function for VaR. It is the training objective for the quantile models of Question 5, the comparison criterion for VaR forecasts in the backtest table (lower is better; it ranks methods but does not test a single one), and the basis of the Fissler–Ziegel score (Appendix A.3).

A.2 Acerbi–Székely Z test

With $(\widehat{\text{VaR}}_{t,\alpha}, \widehat{\text{ES}}_{t,\alpha})$ the forecast VaR/ES pair over n forecast days and exception indicator $I_t = \mathbf{1}\{L_t > \widehat{\text{VaR}}_{t,\alpha}\}$:

$$Z = \frac{1}{n(1-\alpha)} \sum_{t=1}^{n} \frac{L_t I_t}{\widehat{\text{ES}}_{t,\alpha}} - 1.$$

Under correctly specified (continuous) forecast distributions, $\mathbb{E}[Z] = 0$; $Z > 0$ means realized tail losses, scaled by the forecast ES, exceed what the model predicts, through too many exceptions, too large ones, or both. Because Z's sampling distribution has no closed form, simulate it: hold each day's forecast distribution fixed, draw $L_t^{(m)}$ from it independently for every t and $m = 1, \ldots, M$, compute $Z^{(m)}$ with the same forecasts, and report the one-sided p-value $\frac{1}{M}\sum_m \mathbf{1}\{Z^{(m)} \ge Z\}$. For historical simulation, draw by resampling the day's window (its distribution is discrete, so $\mathbb{E}[Z] = 0$ only approximately).

A.3 Fissler–Ziegel joint score and Diebold–Mariano test

With $v = \widehat{\text{VaR}}_{t,\alpha}$, $e = \widehat{\text{ES}}_{t,\alpha} > 0$ and $I_t = \mathbf{1}\{L_t > v\}$:

$$S(v, e, L) = \frac{I_t}{(1-\alpha)e}(L - v) + \frac{v}{e} + \log e - 1.$$

This member of the Fissler and Ziegel (2016) family is due to Patton et al. (2019). S is strictly consistent for the pair $(\text{VaR}_\alpha, \text{ES}_\alpha)$: averaged over time, the correctly specified forecast has the lowest expected score. Rank competing methods by $\bar{S}$; to compare two methods A and B (here or with the pinball loss), test whether the mean score difference $d_t = S_t^A - S_t^B$ is zero (Diebold–Mariano test): a t-test on $\{d_t\}$ with a Newey and West (1987) standard error with 10 lags, since the d_t are autocorrelated (e.g. `statsmodels` OLS of d_t on a constant with `cov_type="HAC"`, `maxlags=10`).

A.4 Information criteria, Ljung–Box and Jarque–Bera tests

Information criteria. For AR(p) with constant variance, fitted by Gaussian quasi-maximum likelihood on the n observations left after conditioning on the first $\bar{p} = 3$, let $\hat{L}$ be the maximized likelihood and $k = p + 2$ the number of estimated parameters ($\phi_1, \ldots, \phi_p$, μ, σ_ε^2). Then

$$AIC = -2\log\hat{L} + 2k, \qquad BIC = -2\log\hat{L} + k\log n,$$

and the candidate with the smallest value is chosen. BIC is consistent (it selects the true order with probability tending to one if it is among the candidates); AIC tends to choose orders that are too large.

Ljung–Box test (Ljung and Box, 1978). For a series $e_1, \dots, e_n$ with sample autocorrelations $\hat{\rho}(j)$,

$$Q_{LB} = n(n+2) \sum_{j=1}^{H} \frac{\hat{\rho}(j)^2}{n-j}.$$

Under H_0 (e_t i.i.d.), the reference distribution is approximately:

- χ^2_H for the returns R_t, or R_t^2 (if $\mathbb{E}[R_t^4] < \infty$);
- χ^2_{H-p} for the residuals $\hat{\varepsilon}_t$ of a fitted AR(p), since p parameters were estimated;
- χ^2_H for the *squared* residuals $\hat{\varepsilon}_t^2$: estimating the AR coefficients does not change their limit distribution (McLeod and Li, 1983);
- χ^2_{H-p} for the standardized GARCH residuals $\hat{z}_t$, and χ^2_{H-2} for $\hat{z}_t^2$, since the two GARCH parameters α_1, β_1 were estimated (an approximation; see Li and Mak, 1994).

Reject at 5% if Q_{LB} exceeds the 95% quantile of that distribution. The approximation relies on the i.i.d. null, so it is not valid under volatility clustering.

Jarque–Bera test. With sample skewness b and sample kurtosis $\hat{\kappa}$ of n observations, $JB = \frac{n}{6}\big(b^2 + \frac{1}{4}(\hat{\kappa} - 3)^2\big) \xrightarrow{d} \chi^2_2$ under H_0: the data are i.i.d. normal.

A.5 GPD tail estimators of VaR and ES

For n losses and a threshold u with N_u exceedances $Y_i = L_i - u > 0$, the excess distribution is approximately GPD(ξ, β) for u large enough (Pickands–Balkema–de Haan). Fitting $(\hat{\xi}, \hat{\beta})$ by maximum likelihood on $\{Y_i\}$, the tail estimate $\hat{\bar{F}}(x) = \frac{N_u}{n}\big(1 + \hat{\xi}\,\frac{x-u}{\hat{\beta}}\big)^{-1/\hat{\xi}}$, $x > u$, gives at a level $\alpha > 1 - N_u/n$ (and $\hat{\xi} \neq 0$)

$$\widehat{\mathrm{VaR}}_\alpha = u + \frac{\hat{\beta}}{\hat{\xi}}\left[\left(\frac{n(1-\alpha)}{N_u}\right)^{-\hat{\xi}} - 1\right], \qquad \widehat{\mathrm{ES}}_\alpha = \frac{\widehat{\mathrm{VaR}}_\alpha}{1-\hat{\xi}} + \frac{\hat{\beta} - \hat{\xi} u}{1-\hat{\xi}} \quad (\hat{\xi} < 1).$$

The ES formula is $\widehat{\mathrm{VaR}}_\alpha + e(\widehat{\mathrm{VaR}}_\alpha)$, with the GPD mean excess function $e(v) = \big(\hat{\beta} + \hat{\xi}(v-u)\big)/(1-\hat{\xi})$, $v \geq u$.